Many Alternatives, Random Incentives: Sharp Bounds and Efficient Estimation in Treatment Selection Models

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Instrumental Variables (IV) with Multiple Treatments

Setting:

- Discrete "instrument" $Z \in \mathcal{Z}$: randomly assigned
- Discrete "treatment" $D \in \mathcal{D}$: generally endogenous
- Outcome(s) $Y \in \mathcal{Y}$ may be continuous: $\mathcal{Y}$ compact subset of $\mathbb{R}^k$

Scope: Models that restrict how individuals select into treatment in response to instruments/incentives/assignment

This Paper: Unified (partial) identification approach

$\rightarrow$ Flexible choice of both target parameter and model

- Exclusion restrictions: natural, but not required

- **Moment inequalities** characterization, sharp for "full" independence

  $\rightarrow$ not just mean independence, can lead to much higher identifying power

- **Estimation and Inference** without discretization

  $\rightarrow$ Substantial gains in statistical power and tractability when Y continuous

Potential Outcomes, Unobservable Heterogeneity

Observables: $Y \in \mathcal{Y}$, $D \in \mathcal{D} := \{d_1, \dots, d_t\}$, $Z \in \mathcal{Z} := \{z_1, \dots, z_m\}$, covariates $X \in \mathcal{X}$

Unobservable Heterogeneity

- Potential outcomes $Y_{d_1}, \dots, Y_{d_t}$ such that $Y = Y_D$,
- Response-type $\boldsymbol{T} := \left(D_{z_1}, \dots, D_{z_m}\right)$ such that $D = D_Z$.

Baseline Assumptions

- $\left(\left(Y_{d_1}, \dots, Y_{d_t}\right), \boldsymbol{T}, Z, X\right) \sim Q_0$ with known support $\mathcal{Y}^t \times \mathcal{D}^m \times \mathcal{Z} \times \mathcal{X}$.
- We observe $(Y, D, Z, X) \sim P$, induced by Q_0 via $Y = Y_D, D = D_Z$.

Additional Assumptions

- Random Assignment: $\left((Y_{d_1}, \dots, Y_{d_t}), \boldsymbol{T}\right) \perp Z \,|\, X$
- Q_0 satisfies $\mathbf{S}$, the *Treatment Selection Model* specified by analyst

$\rightarrow$ In remainder, condition on X and suppress it in the notation

Model Scope: Treatment Selection Models (TSMs)

Introduce unobservables $(\boldsymbol{V}, \varepsilon_Y)$, functions g_D and g_Y:

Treatment Selection Equation: $D = g_D(Z, \boldsymbol{V})$,

Outcome Equation: $Y = g_Y(D, \boldsymbol{V}, \varepsilon_Y)$,

Independence Condition: $Z, \boldsymbol{V}, \varepsilon_Y$ are mutually independent.

Formally equivalent to the PO model, can define

$$Y_d := g_Y(d, \boldsymbol{V}, \varepsilon_Y) \quad \text{and} \quad D_z := g_D(z, \boldsymbol{V}) \ .$$

$$\rightarrow \ \boldsymbol{T} = (g_D(z, \boldsymbol{V}) : z \in \mathcal{Z})$$

Treatment Selection Models (TSMs): Assumptions on treatment selection equation, i.e. restrictions on g_D and/or $\boldsymbol{V}$

- Example: $D = \mathbb{1}\{\boldsymbol{V} \leq \nu(Z)\}$, ν increasing, $\boldsymbol{V} \sim \text{Unif}[0, 1]$, latent index model of Vytlacil (2002)

$\rightarrow$ Such models are characterized by restrictions on *distribution* of $\boldsymbol{T}$

Other Examples of TSMs and Applications

Monotonicity: Imbens & Angrist (1996), Rose & Shem-Tov (2021).

Revealed Preference: Kline & Walters (2016), Kirkebøen, Leuven, & Mogstad (2016), Heckman & Pinto (2018), Pinto (2021).

Encouragement Designs, Generalized Threshold Crossing Models: Heckman and Vytlacil (2001, 2005), Lee and Salanié (2018), Bai, Huang, Tabord-Meehan (2024)

Many IVs: Mogstad, Torgovitsky, & Walters (2020), Mountjoy (2022), Goff (2024), Sun and Wüthrich (2024), Sigstad (2024), van't Hoff et al. (2025).

Mediation and Mechanisms: Imai, Keele, & Yamamoto (2010), Frolich & Huber (2017), Kwon & Roth (2026).

Sample Selection and Non-Response Bias: Lee (2009), Dutz et al. (2026), Kroft, Mourifié, & Vayalinkal (2026).

Overview of Results

Identification

1. **Sharp** testable implications of such models
   - Finite-dimensional convex representation (even if Y continuous)

   $\rightarrow$ Sharp Test for TSMs and IV Validity as **moment inequalities**
2. Partial identification of $\mathbb{E}\left[Y_{d_1} - Y_{d_2} \,\middle|\, \boldsymbol{V} \in \mathcal{V}^*\right]$ and $\frac{\mathbb{E}\left[Y_{d_1} - Y_{d_2} \,\middle|\, \boldsymbol{V} \in \mathcal{V}^*\right]}{\mathbb{E}\left[Y_{d_3} - Y_{d_4} \,\middle|\, \boldsymbol{V} \in \mathcal{V}^*\right]}$
   - **Sharp Bounds** and Parameter-inclusion Tests

$\rightarrow$ Fast, powerful inference via Chernozhukov, Lee and Rosen (2013)

Estimation

1. **Semiparametric efficiency bound** for estimation of sharp bounds
2. Formulate tractable "plug-in" estimator for sharp bounds
3. Propose new cross-fitted bias-corrected estimator for the bounds

   $\rightarrow$ Attains efficiency bound in suitably regular cases

   $\rightarrow$ Mitigates inward bias of "plug-in" estimator in irregular cases

Example: Moving to Opportunity (MTO) Experiment

Let D denote neighbourhood choice. Here, individuals choose from three different types of neighbourhoods: "high" poverty (H), "medium" poverty (M), and "low" poverty (L).

Let $Z = (Z_M, Z_L)$ be the randomly assigned instrument. Individuals living in high poverty neighbourhoods were randomly offered rent vouchers:

$(Z_M, Z_L) = (0,0)$: No vouchers

$(Z_M, Z_L) = (0,1)$: Vouchers for L neighbourhoods

$(Z_M, Z_L) = (1,1)$: Vouchers for M and L neighbourhoods

$\rightarrow$ Therefore, $D \in \{H, M, L\}$ and $Z \in \{(0,0), (0,1), (1,1)\}$.

$\rightarrow$ Response-type: $\boldsymbol{T} = \left(D_{(0,0)}, D_{(0,1)}, D_{(1,1)}\right)$.

Example: Moving to Opportunity (MTO) Experiment

Suppose we impose "Unordered Partial Monotonicity" (Mountjoy, 2022):

- $\rightarrow$ Z_M incentivizes $D = M$ and Z_L incentivizes $D = L$
- $\rightarrow$ Changes in Z don't effect choices among irrelevant alternatives (e.g., $Z : (0,0) \rightarrow (0,1)$ cannot cause $D : H \rightarrow M$)

Shifting $Z : (1,1) \rightarrow (0,1)$ causes two types of moves:

- Individuals with $\boldsymbol{T} \in \{(H,L,M), (M,L,M), (L,L,M)\}$: $M \rightarrow L$
- Individuals with $\boldsymbol{T} = (H,H,M)$: $M \rightarrow H$

The following weighted average can be (point) identified

$$\omega_{M\rightarrow L} E\left[Y_L - Y_M \mid D_{(0,1)} = L, D_{(1,1)} = M\right] + \omega_{M\rightarrow H} E\left[Y_H - Y_M \mid D_{(0,1)} = H, D_{(1,1)} = M\right],$$

but the two component effects are only partially identified.

"Trimming" Bounds of Horowitz and Manski (1995)

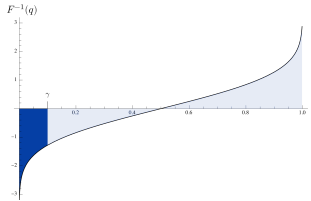
Suppose all we observe is the following mixture distribution, with CDF F,

$$F(y) = \gamma F_1(y) + (1-\gamma) F_2(y) \ ,$$

where $\gamma \in (0,1)$ is known, but F_1 and F_2 are unknown CDFs. Let $Y_1 \sim F_1$.

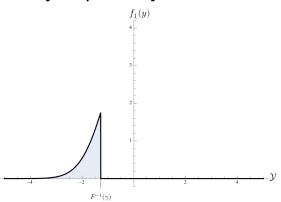
If $\gamma = 0.1$, sharp lower bound for $E[Y_1]$ is constructed as follows:

Lower Bound



(a) Lower bound is area of shaded region

Density Implied by Lower Bound



(b) Density of F_1

Figure 1: Sharp lower bound for $E[Y_1]$, with $\gamma = 0.1$

“Trimming” Bounds of Horowitz and Manski (1995)

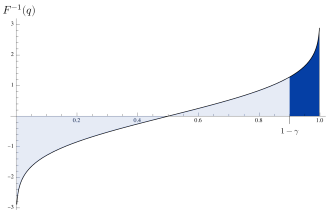
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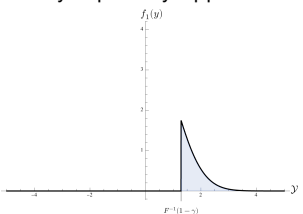
If $\gamma = 0.1$, sharp **upper** bound for $E[Y_1]$ is constructed as follows:

Upper Bound



(a) Upper bound is area of shaded region

Density Implied by Upper Bound



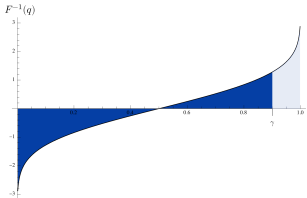
(b) Density of F_1

Figure 1: Sharp upper bound for $E[Y_1]$, with $\gamma = 0.1$

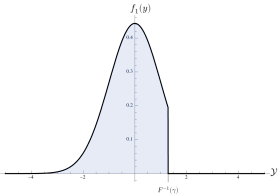
“Trimming” Bounds of Horowitz and Manski (1995)

If instead we have $\gamma = 0.9$, then lower bound is

Lower Bound



Density Implied by Lower Bound



(a) Lower bound is area of shaded region

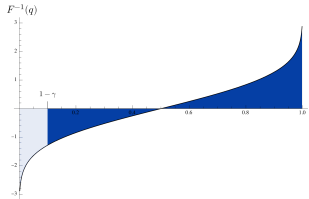
(b) Density of F_1

Figure 2: Sharp lower bound for $E[Y_1]$, with $\gamma = 0.9$

“Trimming” Bounds of Horowitz and Manski (1995)

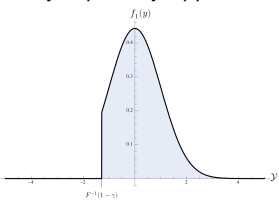
and upper bound is

Upper Bound



(a) Upper bound is area of shaded region

Density Implied by Upper Bound



(b) Density of F_1

Figure 2: Sharp upper bound for $E[Y_1]$, with $\gamma = 0.9$

Horowitz and Manski (1995) Bounds: Takeaway

The observed mixture distribution, written in terms of densities, is

$$f(y) = \gamma f_1(y) + (1-\gamma) f_2(y),$$

where f_1 is the density of F_1, and f_2 is the density of F_2.

This implies two restrictions on the function $\gamma f_1(\cdot)$:

- → point-wise restriction: $0 \leq \gamma f_1(y) \leq f(y)$, for all y
- → mass restriction: $\int_{-\infty}^{\infty} f_1(y)\, d\mu(y) = 1 \implies \int_{-\infty}^{\infty} \gamma f_1(y)\, d\mu(y) = \gamma$

Note that $E[Y_1] := \frac{1}{\gamma} \int_{-\infty}^{\infty} y \gamma f_1(y)\ d\mu(y)$

- → optimizing the right hand side over all functions $\gamma f_1(\cdot)$ that satisfy the two conditions above results in the Horowitz-Manski bounds

IV Mixture Model: Basic Idea

For any $d \in \mathcal{D}$ and each $z \in \mathcal{Z}$,

We observe Y for those assigned $Z = z$ and took treatment d

We want to learn about Y_d for those of type $\boldsymbol{T} = \boldsymbol{\tau}$, for some $\boldsymbol{\tau} \in \mathcal{T}$

Key: Individuals who take treatment $D = d$ when $Z = z$ consists of the mixture of individuals of types $\boldsymbol{T} = \boldsymbol{\tau}$ such that $\boldsymbol{\tau}(z) = d$.

$\rightarrow$ Observed outcome distributions are mixture distributions:

$$\sum_{\boldsymbol{\tau} \in \mathcal{T}} \underbrace{\mathbb{1}\{\boldsymbol{\tau}(z) = d\}}_{\text{known}} \underbrace{\mathbb{P}_{Q_0}(\boldsymbol{T} = \boldsymbol{\tau})\, \mathbb{P}_{Q_0}(Y_d \leq y \mid \boldsymbol{T} = \boldsymbol{\tau})}_{\text{unobserved}}$$

$$= \underbrace{\mathbb{P}_P(D = d \mid Z = z)\, \mathbb{P}_P(Y \leq y \mid D = d, Z = z)}_{\text{observed}},$$

$\rightarrow$ Allowing Z to vary, get linear system with known coefficients

IV Mixture Model: Informal Overview of Approach

1. From model $\mathbf{S}$, get linear system coeffs., via response-matrices $\mathbf{R}_d$,

$$\mathbf{R}_d \boldsymbol{q}_d(y) = \boldsymbol{p}_d(y) \quad \text{and} \quad \boldsymbol{q}_d(y) \geq 0 \qquad (\star)$$

$\rightarrow$ $\boldsymbol{p}_d$: observed densities $\boldsymbol{p}_{d|Z=z} = \mathbb{P}_P(D = d|Z = z) f_{Y|D=d,Z=z}$

$\rightarrow$ $\boldsymbol{q}_d$ unobserved type-conditional densities $\mathbb{P}_{Q_0}(\boldsymbol{T} = \boldsymbol{\tau})\, f_{Y_d|\boldsymbol{T}=\boldsymbol{\tau}}$

Basic Example ($Z \in \{0,1\}$, $D \in \{0,1\}$, unrestricted selection)

In this case, the response types are always-takers (AT), compliers (C), defiers (D), never-takers (NT) , leading to

$$\mathbf{R}_0 = \begin{array}{c} \begin{matrix} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{matrix} \\ \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array} \begin{matrix} (Z=0) \\ (Z=1) \end{matrix} \quad \text{and} \quad \mathbf{R}_1 = \begin{array}{c} \begin{matrix} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{matrix} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

$q_d(y)$ must be a solution to $(\star)$

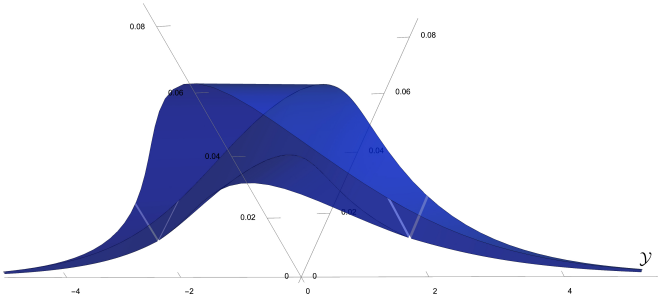


Figure 3: Hypothetical graph of the correspondence $y \mapsto \tilde{\mathcal{B}}_d\left(\boldsymbol{p}_d(y)\right)$

$\boldsymbol{q}_d(y)$ must be a solution to $(\star)$

$\rightarrow$ If $\mathbf{x}$ is a vector-valued function such that $\mathbf{x}(y) \in \mathcal{B}_d(\boldsymbol{p}_d(y))$ for all $y \in \mathcal{Y}$, then $\mathbf{x}$ is a "candidate" for $\boldsymbol{q}_d(y)$.

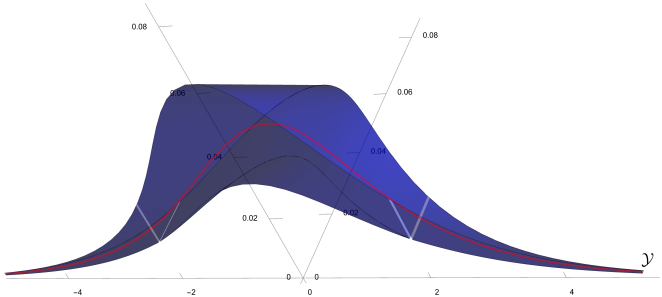


Figure 3: Hypothetical graph of the correspondence $y \mapsto \tilde{\mathcal{B}}_d(\boldsymbol{p}_d(y))$

IV Mixture Model: Informal Overview of Approach

2. Each solution $\boldsymbol{q}_d$ of $(\star)$ integrates to a "candidate" value of

$$\boldsymbol{\pi} := \left(\mathbb{P}_{Q_0}\left(\boldsymbol{T} = \boldsymbol{\tau}\right)\right)_{\boldsymbol{\tau} \in \{\text{types possible under } \mathbf{S}\}}$$

For each d, call Π_d the set of such candidate values.

$\rightarrow$ If $\mathbf{S}$ simply eliminates types, then $\mathbf{S}$ rejected $\iff \bigcap_{d \in \mathcal{D}} \Pi_d = \emptyset$!

Basic Example ($Z \in \{0,1\}$, $D \in \{0,1\}$, unrestricted selection)

Since $\mathbf{R}_1 = \begin{array}{c} \begin{matrix} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{matrix} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \end{array}$ and $\boldsymbol{q}_d \geq 0$,

$$\mathbf{R}_1 \boldsymbol{q}_d(y) = \boldsymbol{p}_d(y) \implies \boldsymbol{q}_{d|\boldsymbol{T}=\text{AT}}(y) \leq \min\left\{\boldsymbol{p}_{1|Z=0}(y), \boldsymbol{p}_{1|Z=1}(y)\right\}$$

$$\implies \boldsymbol{\pi}_{(\text{AT})} \leq \int_{\mathcal{Y}} \min\left\{\boldsymbol{p}_{1|Z=0}(y), \boldsymbol{p}_{1|Z=1}(y)\right\} dy$$

Example 1: Head Start Impact Study

Let D denote preschool choice. Families choose between no preschool (n), competing preschools (c), and Head Start (h) for their child.

The Head Start Impact Study randomized applicants into two groups:

- $Z = 0$: Not offered admission to Head Start
- $Z = 1$: Offered admission to Head Start.

$\rightarrow$ Therefore, $D \in \{n, c, h\}$ and $Z \in \{0, 1\}$.

$\rightarrow$ Response-type: $\boldsymbol{T} = (D_0, D_1)$.

Example 1 (Head Start): Partial Identification

Assume that the following assumption holds for everyone

- either $D_1 = D_0$ or $\mathbb{1}\{D_1 = h\} > \mathbb{1}\{D_0 = h\}$

Shifting $Z : 0 \to 1$ causes two types of enrolment changes:

- Individuals with $\boldsymbol{T} = (n, h)$: $n \to h$
- Individuals with $\boldsymbol{T} = (c, h)$: $c \to h$

The following weighted average can be (point) identified

$$\omega_{n\to h} E\left[Y_h - Y_n \mid D_0 = n, D_1 = h\right] + \omega_{c\to h} E\left[Y_h - Y_c \mid D_0 = c, D_1 = h\right] ,$$

but the two component effects are only partially identified.

Example 1 (Head Start): "h"-response-type

Consider the effect $E\left[Y_h - Y_c \mid \boldsymbol{T} = (ch)\right]$

$\rightarrow$ $P\left[\boldsymbol{T} = (ch)\right]$ and $E\left[Y_c \mid \boldsymbol{T} = (ch)\right]$ are point-identified

$\rightarrow$ Therefore, it suffices to find the bounds for

$$P\left[\boldsymbol{T} = (ch)\right] E\left[Y_h \mid \boldsymbol{T} = (ch)\right]$$

There are two types such that $D_0 \neq h$ and $D_1 = h$:

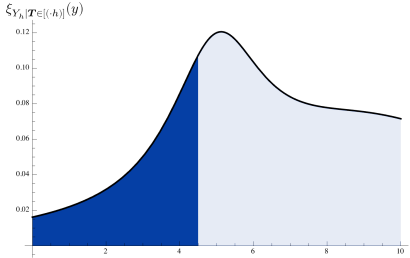
$$\{(ch), (nh)\}$$

Aggregate $Y_h | \boldsymbol{T} \in \{(ch), (nh)\}$ is point-identified.

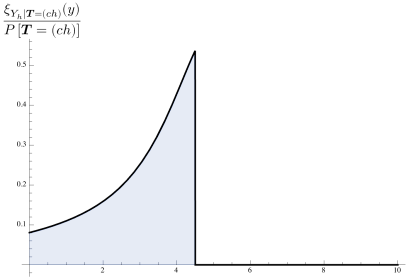
$\rightarrow$ Trim the identified density $P\left[\boldsymbol{T} \in \{(ch), (nh)\}\right] f_{Y_h | \boldsymbol{T} \in \{(ch),(nh)\}}(y)$

Example 1 (Head Start): Sharp Lower Bound

Suppose $P[\boldsymbol{T} = (ch)] = 0.2$.



(a) Dark blue region has area 0.2.



(b) Density $\frac{q_{Y_h|\boldsymbol{T}=ch)}}{P[\boldsymbol{T}=(ch)]} = f_{Y_h|\boldsymbol{T}=(ch)}$ implied by sharp lower bound

Figure 4: (Simulated) Construction of sharp lower bound for $E[Y_h|\boldsymbol{T} = (ch)]$

Example 2 (MTO): "M"-response-type

Recall: $D \in \{H, M, L\}$ and $Z = (Z_M, Z_L) \in \{(0,0), (0,1), (1,1)\}$.
Response-type: $\boldsymbol{T} = \left(D_{(0,0)} D_{(0,1)} D_{(1,1)}\right)$.

Consider the effect $E\left[Y_H - Y_M \mid \boldsymbol{T} = (HHM)\right]$

$\rightarrow$ $P\left[\boldsymbol{T} = (HHM)\right]$ and $E\left[Y_H \mid \boldsymbol{T} = (HHM)\right]$ are point-identified

$\rightarrow$ Suffices to find the bounds for

$$P\left[\boldsymbol{T} = (HHM)\right] E\left[Y_M \mid \boldsymbol{T} = (HHM)\right]$$

There are three types such that $D_{(0,0)} \neq M, D_{(0,1)} \neq M$ and $D_{(1,1)} = M$:

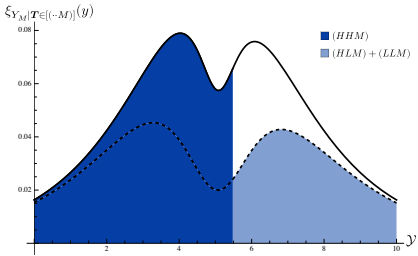
$$\{(HHM), (HLM), (LLM)\}$$

Dist. of Y_M for these types is not identified, but can be bounded

Example 2 (MTO): Non-Sharpness of Simple Bounds

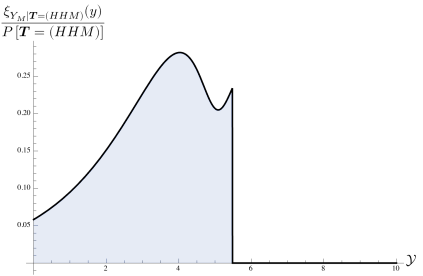
Suppose the sharp upper bound on $P[\boldsymbol{T} \in [(\cdot\cdot M)]]$ is 0.36, and $P[\boldsymbol{T} = (HHM)] = 0.28$.

Construction of Simple Lower Bound



dark blue area $= 0.28$
dark blue area + light blue area $= \mathbf{0.423}$

Density $f_{Y_M|\boldsymbol{T}=(HHM)}$ Implied by Simple Lower Bound



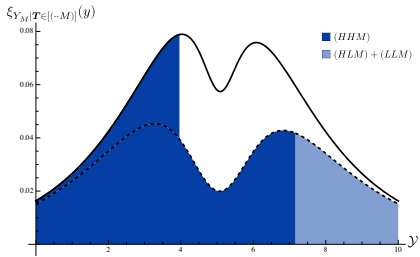
This density implies the simple lower bound of 3.289

Figure 5: (Simulated) Construction of lower bounds for $E[Y_M|\boldsymbol{T} = (HHM)]$

Example 2 (MTO): Non-Sharpness of Simple Bounds

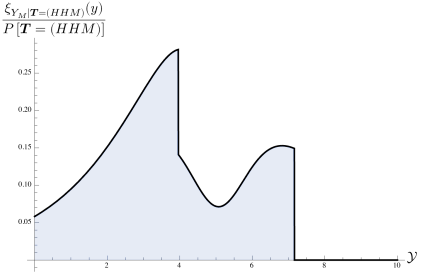
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Construction of Sharp Lower Bound



dark blue area $= 0.28$
dark blue area + light blue area $= \mathbf{0.36}$

Density $f_{Y_M|\boldsymbol{T}=(HHM)}$ Implied by Sharp Lower Bound



This density implies the sharp lower bound of 3.665

Figure 5: (Simulated) Construction of lower bounds for $E[Y_M|\boldsymbol{T} = (HHM)]$

Example 3: Multi-dimensional Treatment

Consider a setting with a primary treatment, $P \in \{0,1\}$, and supplementary treatment, $S \in \{0,1\}$, and define $D = (P,S)$.

Some individuals are randomly offered incentives to participate in treatment. Let $Z = (Z_P, Z_S)$ where

- $Z_P \in \{0,1\}$ represents an incentive to participate in P
- $Z_S \in \{0,1\}$ represents an additional incentive to participate in S

Suppose individuals are

- only offered supplementary incentive if offered primary incentive
- only allowed supplementary treatment if they take primary treatment

$\rightarrow$ Therefore, $D \in \{(0,0),(1,0),(1,1)\}$ and $Z \in \{(0,0),(1,0),(1,1)\}$.

$\rightarrow$ Response-type: $\boldsymbol{T} = \left(D_{(0,0)}, D_{(1,0)}, D_{(1,1)}\right)$.

Multi-dimensional Treatment Example

Assume that the following holds for all individuals:

1. $Z_P : 0 \to 1$ (weakly) increases $\mathbb{1}\{D \in \{(1,0),(1,1)\}\}$ and $\mathbb{1}\{S = 1\}$,
2. $Z_S : 0 \to 1$ (weakly) increases $\mathbb{1}\{D = (1,1)\}$ and $\mathbb{1}\{P = 1\}$.

Many potential parameters of interest...

For example, consider the decomposition

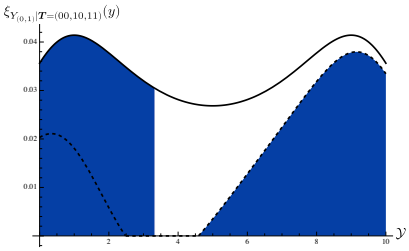
$$\underbrace{E\left[Y_{(1,1)} - Y_{(0,0)} \mid \boldsymbol{T} = ((0,0),(1,0),(1,1))\right]}_{\text{Total Effect}}$$
$$= \underbrace{E\left[Y_{(1,1)} - Y_{(1,0)} \mid \boldsymbol{T} = ((0,0),(1,0),(1,1))\right]}_{\text{Supplementary Effect}}$$
$$+ \underbrace{E\left[Y_{(1,0)} - Y_{(0,0)} \mid \boldsymbol{T} = ((0,0),(1,0),(1,1))\right]}_{\text{Primary Effect}}$$

$\to$ **All components are only partially identified!**

Multi-dim. Treatment example: Tighter Bounds due to Pointwise Dependence

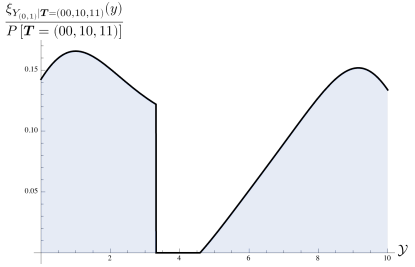
Suppose $P\left[\boldsymbol{T} = (00, 10, 11)\right] = 0.25$.

Construction of Simple Lower Bound



dark blue area $= 0.25$

Density $f_{Y_{(1,0)}|\boldsymbol{T}=(00,10,11)}$ Implied by Simple Lower Bound



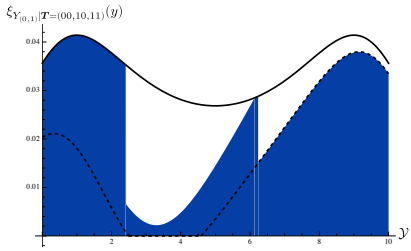
This density implies the simple lower bound of 4.865

Figure 6: Construction of lower bounds for $E\left[Y_{(1,0)}|\boldsymbol{T} = (00, 10, 11)\right]$

Multi-dim. Treatment example: Tighter Bounds due to Pointwise Dependence

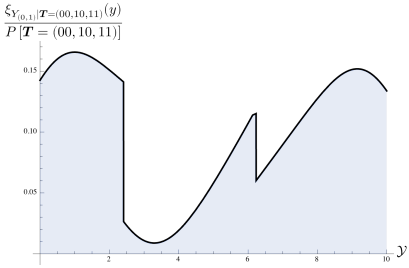
Suppose $P\left[\boldsymbol{T} = (00, 10, 11)\right] = 0.25$.

Construction of Sharp Lower Bound



dark blue area $= 0.25$

Density $f_{Y_{(1,0)}|\boldsymbol{T}=(00,10,11)}$ Implied by Sharp Lower Bound



This density implies the sharp lower bound of 5.106

Figure 6: Construction of lower bounds for $E\left[Y_{(1,0)}|\boldsymbol{T} = (00, 10, 11)\right]$

Estimating the Bounds

Generally, sharp upper bound $\bar{\theta}(P)$ defined by $\overline{\Psi}\left(\bar{\theta}(P); P\right) = 0$

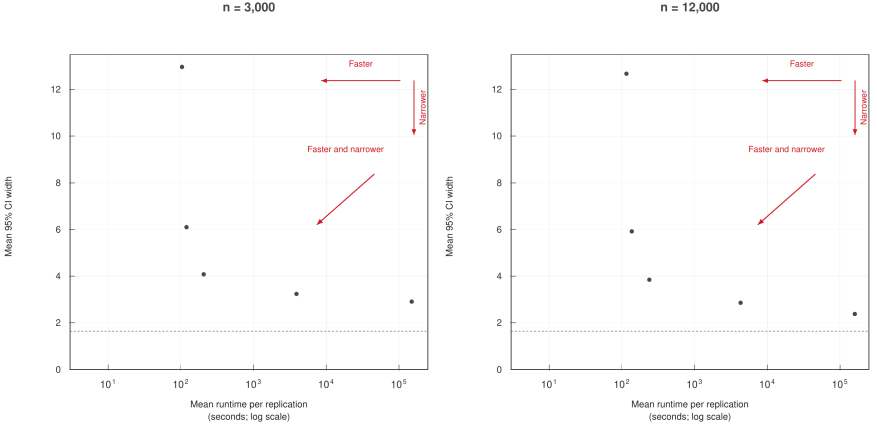
Where $\gamma \mapsto \overline{\Psi}(\gamma; P)$ is a decreasing function of the form

$$\overline{\Psi}(\gamma; P) := \inf_{\boldsymbol{u} \in <\text{known subset of } \mathbb{R}^p>} \sum_{z \in \mathcal{Z}} \mathbb{E}_P \left[\boldsymbol{m}_\theta^P(\gamma, \boldsymbol{u}; Y, D, Z) \,\middle|\, Z = z \right] .$$

- → In practice, don't know P, so can "plug-in" estimate $\tilde{P}_n$
- → Infimum of estimated objective $\implies \mathbb{E}_P\left[\overline{\Psi}\left(\gamma; \tilde{P}_n\right)\right] < \overline{\Psi}(\gamma; P)$
- → $\overline{\Psi}\left(\bar{\theta}(P); P\right) = 0 = \overline{\Psi}\left(\bar{\theta}\left(\tilde{P}_n\right); \tilde{P}\right) < \overline{\Psi}\left(\bar{\theta}\left(\tilde{P}_n\right); P\right)$
- → $\implies \bar{\theta}\left(\tilde{P}_n\right) < \bar{\theta}(P)$, since $\overline{\Psi}$ decreasing
- → **In paper:** bias-corrected estimator!

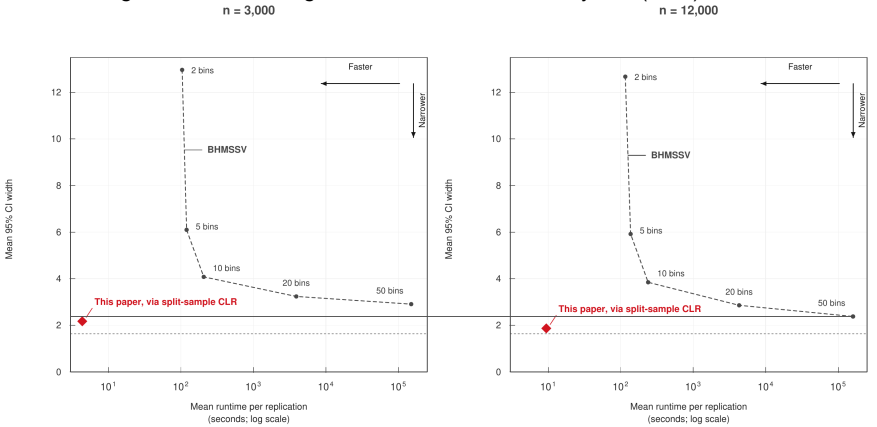
Inference: Comparison to Discretized Methods

DGP from Figure 1 of Bai, Huang, Moon, Santos, Shaikh, & Vytlacil (2026)



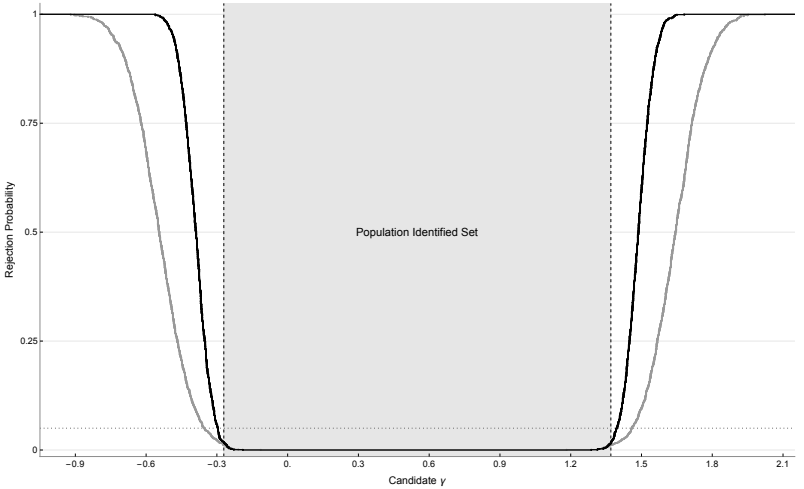
Inference: Comparison to Discretized Methods

DGP from Figure 1 of Bai, Huang, Moon, Santos, Shaikh, & Vytlacil (2026)



Inference: Comparison to Discretized Methods

Inference: Power Curves



Grey = 3k, Black = 12k; 1300 Samples.

Works Cited i

Horowitz, J. L. and C. F. Manski (1995). Identification and Robustness with Contaminated and Corrupted Data. *Econometrica 63*(2), 281–302.

Mountjoy, J. (2022, August). Community Colleges and Upward Mobility. *American Economic Review 112*(8), 2580–2630.